Problem 4.29: The Systems-of-Equations Method

Build the decompositions, then invert them to get product-state form

CG coefficients from tables only — no angular momentum algebra

The Problem

Find the ratio of cross sections for:

(a) $\pi^- + p \to K^0 + \Sigma^0$

(b) $\pi^- + p \to K^+ + \Sigma^-$

(c) $\pi^+ + p \to K^+ + \Sigma^+$

First when $I = 3/2$ dominates, then when $I = 1/2$ dominates.

The game plan

For each reaction we will:

1. Write out the reaction
2. Work the initial state: find I and I_3 for each particle, write their kets, use the triangle rule and I_3 conservation to find which total-isospin kets we need, then decompose those kets using the CG table, and finally **invert** (using systems of equations) to express the reaction's product state in terms of total-isospin kets
3. Do the same for the final state
4. Read off coefficients and compute the amplitude

Reaction (a): $\pi^- + p \to K^0 + \Sigma^0$

Step 1: Write Out the Reaction

$$\pi^- + p \longrightarrow K^0 + \Sigma^0 \tag{1}$$

We will work the **initial state** first, then the **final state**.

Initial State: $\pi^- + p$

Step 2: Find I and I_3 for each particle

Proton ($p = uud$):

- I_3: add quarks: $+1/2 + 1/2 + (-1/2) = +1/2$
- I: proton is part of a doublet (p, n), so $2I + 1 = 2 \Rightarrow I = 1/2$

$$p = \left|\frac{1}{2}, +\frac{1}{2}\right\rangle \tag{2}$$

Pion ($\pi^- = d\bar{u}$):

- I_3: add quarks: $(-1/2) + (-1/2) = -1$
- I: π^- is part of a triplet (π^+, π^0, π^-), so $2I + 1 = 3 \Rightarrow I = 1$.
  Alternatively: $I_3 = -1$ can only fit in $I = 1$ (stretched state argument).

$$\pi^- = |1, -1\rangle \tag{3}$$

Step 3: Build the total-isospin kets we need

Triangle rule for $\pi^- + p$: combining $I_\pi = 1$ and $I_p = 1/2$:

$$|I_\pi - I_p| \leq I \leq I_\pi + I_p \quad \Longrightarrow \quad \frac{1}{2} \leq I \leq \frac{3}{2} \tag{4}$$

So $I = 1/2$ or $I = 3/2$.
I_3 **conservation**: $I_3^{\text{total}} = I_{3,\pi^-} + I_{3,p} = (-1) + (+1/2) = -1/2$.
Which total-isospin states have $I_3 = -1/2$?

- $I = 3/2$: allowed $I_3 = -3/2, -1/2, +1/2, +3/2$. **Yes**, $-1/2$ fits.
- $I = 1/2$: allowed $I_3 = -1/2, +1/2$. **Yes**, $-1/2$ fits.

So we need these two kets:

$$\left|\frac{3}{2}, -\frac{1}{2}\right\rangle \quad \text{and} \quad \left|\frac{1}{2}, -\frac{1}{2}\right\rangle \tag{5}$$

Step 4: Decompose these kets using the CG table

We use the $1 \times 1/2$ CG table (pion first, since $I_\pi = 1$ is the bigger isospin).
First: which product states $|m_1, m_2\rangle$ have $m_1 + m_2 = -1/2$?

- $m_1 = 0,\ m_2 = -1/2$ (π^0 and n) ✓
- $m_1 = -1,\ m_2 = +1/2$ (π^- and p) ✓
- $m_1 = -1/2,\ m_2 = 0$? No! Pion can't have $m_1 = -1/2$ (it has $I = 1$, so m_1

must be integer).

Only two product states survive.

Decompose $|3/2, -1/2\rangle$: Go to the CG table, read the **row** for $|3/2, -1/2\rangle$. Pick up the entries under each surviving column:

Column (m_1, m_2)	**Table entry**	**Coefficient (take $\sqrt{\ }$)**
$m_1 = 0,\ m_2 = -1/2$	$2/3$	$\sqrt{2/3}$
$m_1 = -1,\ m_2 = +1/2$	$1/3$	$\sqrt{1/3}$

$$\left|\frac{3}{2}, -\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\,|0, -\tfrac{1}{2}\rangle + \sqrt{\frac{1}{3}}\,|-1, +\tfrac{1}{2}\rangle \tag{6}$$

In particle language: $|0, -1/2\rangle = |\pi^0\rangle|n\rangle$ and $|-1, +1/2\rangle = |\pi^-\rangle|p\rangle$.

$$\left|\frac{3}{2}, -\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\,|\pi^0\rangle|n\rangle + \sqrt{\frac{1}{3}}\,|\pi^-\rangle|p\rangle \tag{Eq. 1}$$

Decompose $|1/2, -1/2\rangle$: Same columns, read the $|1/2, -1/2\rangle$ row:

Column (m_1, m_2)	**Table entry**	**Coefficient**
$m_1 = 0,\ m_2 = -1/2$	$1/3$	$\sqrt{1/3}$
$m_1 = -1,\ m_2 = +1/2$	$-2/3$	$-\sqrt{2/3}$

$$\left|\frac{1}{2}, -\frac{1}{2}\right\rangle = \sqrt{\frac{1}{3}}\,|\pi^0\rangle|n\rangle - \sqrt{\frac{2}{3}}\,|\pi^-\rangle|p\rangle \tag{Eq. 2}$$

Quick checks on Eq. 1 and Eq. 2

Normalization:

- Eq. 1: $(\sqrt{2/3})^2 + (\sqrt{1/3})^2 = 2/3 + 1/3 = 1$ ✓
- Eq. 2: $(\sqrt{1/3})^2 + (-\sqrt{2/3})^2 = 1/3 + 2/3 = 1$ ✓

Orthogonality:

$$\sqrt{\frac{2}{3}} \cdot \sqrt{\frac{1}{3}} + \sqrt{\frac{1}{3}} \cdot \left(-\sqrt{\frac{2}{3}}\right) = \frac{\sqrt{2}}{3} - \frac{\sqrt{2}}{3} = 0 \quad \checkmark \tag{7}$$

Step 5: Invert using systems of equations to get $|\pi^- p\rangle$

We have two equations (Eq. 1 and Eq. 2) and we want to isolate $|\pi^-\rangle|p\rangle = |-1, +1/2\rangle$. This is a system of two equations with two unknowns ($|\pi^0 n\rangle$ and $|\pi^- p\rangle$).

Goal: eliminate $|\pi^0\rangle|n\rangle$ and solve for $|\pi^-\rangle|p\rangle$.

Multiply Eq. 1 by $\sqrt{1/3}$:

$$\sqrt{\frac{1}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle = \sqrt{\frac{1}{3}}\cdot\sqrt{\frac{2}{3}}\,|\pi^0 n\rangle + \sqrt{\frac{1}{3}}\cdot\sqrt{\frac{1}{3}}\,|\pi^- p\rangle = \frac{\sqrt{2}}{3}\,|\pi^0 n\rangle + \frac{1}{3}\,|\pi^- p\rangle \quad (1')$$

Multiply Eq. 2 by $-\sqrt{2/3}$:

$$-\sqrt{\frac{2}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle = -\sqrt{\frac{2}{3}}\cdot\sqrt{\frac{1}{3}}\,|\pi^0 n\rangle + \sqrt{\frac{2}{3}}\cdot\sqrt{\frac{2}{3}}\,|\pi^- p\rangle = -\frac{\sqrt{2}}{3}\,|\pi^0 n\rangle + \frac{2}{3}\,|\pi^- p\rangle \quad (2')$$

Add (1′) and (2′): The $|\pi^0 n\rangle$ terms cancel!

$$\begin{aligned}\sqrt{\frac{1}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle &= \left(\frac{1}{3} + \frac{2}{3}\right)|\pi^- p\rangle \\ &= 1\cdot|\pi^- p\rangle \end{aligned} \quad (8)$$

Therefore:

$$\boxed{|\pi^- p\rangle = \sqrt{\frac{1}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle} \quad \text{(Initial)}$$

Why did this work?

The CG decomposition (Eq. 1 and 2) expressed total-isospin kets in terms of product kets. We wanted the reverse: express a product ket in terms of total-isospin kets. Since the CG coefficients form an orthogonal matrix, you can always invert by solving a system of equations. The multipliers ($\sqrt{1/3}$ and $-\sqrt{2/3}$) were chosen to cancel the unwanted $|\pi^0 n\rangle$ term.

Shortcut: you could also just read down the **column** $m_1 = -1, m_2 = +1/2$ of the CG table. That column directly gives you the coefficients of $|3/2, -1/2\rangle$ and $|1/2, -1/2\rangle$ in the product state $|-1, +1/2\rangle$. The systems-of-equations method and reading the column are equivalent!

Final State: $K^0 + \Sigma^0$

Step 2: Find I and I_3 for each particle

Kaon ($K^0 = d\bar{s}$):

- $I_3 = (-1/2) + 0 = -1/2$ (the $\bar{s}$ has $I = 0$)
- Only one light quark, so $I = 1/2$ automatically.

$$K^0 = \left|\frac{1}{2}, -\frac{1}{2}\right\rangle \tag{9}$$

Sigma ($\Sigma^0 = uds$):

- $I_3 = (+1/2) + (-1/2) + 0 = 0$
- Part of a triplet $(\Sigma^+, \Sigma^0, \Sigma^-)$, so $2I + 1 = 3 \Rightarrow I = 1$.

$$\Sigma^0 = |1,\, 0\rangle \tag{10}$$

The direct-product ket for the final state is:

$$|K^0\rangle|\Sigma^0\rangle = \left|\frac{1}{2}, -\frac{1}{2}\right\rangle |1, 0\rangle \tag{11}$$

Step 3: Build the total-isospin kets we need

Triangle rule for $K^0 + \Sigma^0$: combining $I_K = 1/2$ and $I_\Sigma = 1$:

$$|I_K - I_\Sigma| \leq I \leq I_K + I_\Sigma \quad \Longrightarrow \quad \frac{1}{2} \leq I \leq \frac{3}{2} \tag{12}$$

So $I = 1/2$ or $I = 3/2$.
I_3 **conservation**: $I_{3,K^0} + I_{3,\Sigma^0} = (-1/2) + 0 = -1/2$.
Same as the initial state: $I_3 = -1/2$. Both $I = 3/2$ and $I = 1/2$ can have $I_3 = -1/2$.
We need the same two kets: $|3/2, -1/2\rangle$ and $|1/2, -1/2\rangle$.

Step 4: Decompose these kets using the CG table

Coupling order issue

The CG table is printed for $1 \times 1/2$ (bigger isospin first). But our final state has the kaon ($I = 1/2$) and sigma ($I = 1$), so if we put the kaon first we need $1/2 \times 1$.

The fix: Look up the $1 \times 1/2$ table with sigma as particle 1 and kaon as particle 2, then apply the coupling-order phase:

$$\text{Phase} = (-1)^{j_1+j_2-J} = (-1)^{1+1/2-J} \tag{13}$$

- $J = 3/2$: phase $= (-1)^0 = +1$ (keep coefficient as-is)
- $J = 1/2$: phase $= (-1)^1 = -1$ (flip sign)

Which product states have $m_K + m_\Sigma = -1/2$? Writing as $|m_\Sigma, m_K\rangle$ (sigma first for the table):

- $m_\Sigma = 0,\ m_K = -1/2 \quad (\Sigma^0 \text{ and } K^0)$ ✓
- $m_\Sigma = -1,\ m_K = +1/2 \quad (\Sigma^- \text{ and } K^+)$ ✓

Decompose $|3/2, -1/2\rangle$: Read the row from the $1 \times 1/2$ table, then apply the phase:

Column	Table entry	Phase ($J = 3/2$)	Final coeff.
$m_\Sigma = 0,\ m_K = -1/2$	$\sqrt{2/3}$	$\times(+1)$	$\sqrt{2/3}$
$m_\Sigma = -1,\ m_K = +1/2$	$\sqrt{1/3}$	$\times(+1)$	$\sqrt{1/3}$

$$\left|\frac{3}{2}, -\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\,|K^0\rangle|\Sigma^0\rangle + \sqrt{\frac{1}{3}}\,|K^+\rangle|\Sigma^-\rangle \tag{Eq. 3}$$

Decompose $|1/2, -1/2\rangle$: Same columns, but now $J = 1/2$ so the phase is (-1):

Column	Table entry	Phase ($J = 1/2$)	Final coeff.
$m_\Sigma = 0,\ m_K = -1/2$	$\sqrt{1/3}$	$\times(-1)$	$-\sqrt{1/3}$
$m_\Sigma = -1,\ m_K = +1/2$	$-\sqrt{2/3}$	$\times(-1)$	$+\sqrt{2/3}$

$$\left|\frac{1}{2}, -\frac{1}{2}\right\rangle = -\sqrt{\frac{1}{3}}\,|K^0\rangle|\Sigma^0\rangle + \sqrt{\frac{2}{3}}\,|K^+\rangle|\Sigma^-\rangle \tag{Eq. 4}$$

Quick checks on Eq. 3 and Eq. 4

Normalization:

- Eq. 3: $2/3 + 1/3 = 1$ ✓
- Eq. 4: $1/3 + 2/3 = 1$ ✓

Orthogonality:

$$\sqrt{\frac{2}{3}} \cdot \left(-\sqrt{\frac{1}{3}}\right) + \sqrt{\frac{1}{3}} \cdot \sqrt{\frac{2}{3}} = -\frac{\sqrt{2}}{3} + \frac{\sqrt{2}}{3} = 0 \quad \checkmark \tag{14}$$

Step 5: Invert using systems of equations to get $|K^0\Sigma^0\rangle$

We have Eq. 3 and Eq. 4, and we want to isolate $|K^0\rangle|\Sigma^0\rangle$. We need to eliminate $|K^+\rangle|\Sigma^-\rangle$.

Multiply Eq. 3 by $\sqrt{2/3}$:

$$\sqrt{\frac{2}{3}} \left|\frac{3}{2}, -\frac{1}{2}\right\rangle = \frac{2}{3} |K^0\Sigma^0\rangle + \frac{\sqrt{2}}{3} |K^+\Sigma^-\rangle \tag{3′}$$

Multiply Eq. 4 by $-\sqrt{1/3}$:

$$-\sqrt{\frac{1}{3}} \left|\frac{1}{2}, -\frac{1}{2}\right\rangle = \frac{1}{3} |K^0\Sigma^0\rangle - \frac{\sqrt{2}}{3} |K^+\Sigma^-\rangle \tag{4′}$$

Add (3′) and (4′): The $|K^+\Sigma^-\rangle$ terms cancel!

$$\begin{aligned} \sqrt{\frac{2}{3}} \left|\frac{3}{2}, -\frac{1}{2}\right\rangle - \sqrt{\frac{1}{3}} \left|\frac{1}{2}, -\frac{1}{2}\right\rangle &= \left(\frac{2}{3} + \frac{1}{3}\right) |K^0\Sigma^0\rangle \\ &= 1 \cdot |K^0\Sigma^0\rangle \end{aligned} \tag{15}$$

Therefore:

$$\boxed{|K^0\Sigma^0\rangle = \sqrt{\frac{2}{3}} \left|\frac{3}{2}, -\frac{1}{2}\right\rangle - \sqrt{\frac{1}{3}} \left|\frac{1}{2}, -\frac{1}{2}\right\rangle} \tag{Final-a}$$

Amplitude for Reaction (a)

Step 6: Compute the amplitude

We have:

$$|\pi^- p\rangle = \sqrt{\frac{1}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle \qquad \text{(Initial)}$$

$$|K^0\Sigma^0\rangle = \sqrt{\frac{2}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle - \sqrt{\frac{1}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle \qquad \text{(Final-a)}$$

The amplitude is: multiply the coefficient of each $|I, I_3\rangle$ in the initial state by the corresponding coefficient in the final state, then multiply by $\mathcal{A}_I$, and add.

Channel	**Initial coeff.**	**Final coeff.**	**Product**
$I = 3/2$	$\sqrt{1/3}$	$\sqrt{2/3}$	$\sqrt{1/3} \times \sqrt{2/3} = \dfrac{\sqrt{2}}{3}$
$I = 1/2$	$-\sqrt{2/3}$	$-\sqrt{1/3}$	$(-\sqrt{2/3}) \times (-\sqrt{1/3}) = \dfrac{\sqrt{2}}{3}$

$$\mathcal{M}_a = \frac{\sqrt{2}}{3}\,\mathcal{A}_{3/2} + \frac{\sqrt{2}}{3}\,\mathcal{A}_{1/2} \tag{16}$$

$$\boxed{\mathcal{M}_a = \frac{\sqrt{2}}{3}\left(\mathcal{A}_{3/2} + \mathcal{A}_{1/2}\right)} \tag{17}$$

$$|\mathcal{M}_a|^2 = \frac{2}{9}\left|\mathcal{A}_{3/2} + \mathcal{A}_{1/2}\right|^2 \tag{18}$$

Reaction (b): $\pi^- + p \to K^+ + \Sigma^-$

Step 1: Write Out the Reaction

$$\pi^- + p \longrightarrow K^+ + \Sigma^- \tag{19}$$

The initial state is the same as reaction (a) — we already have it!

$$|\pi^- p\rangle = \sqrt{\frac{1}{3}}\left|\frac{3}{2}, -\frac{1}{2}\right\rangle - \sqrt{\frac{2}{3}}\left|\frac{1}{2}, -\frac{1}{2}\right\rangle \qquad \text{(Initial — same)}$$

We just need the final state.

Final State: $K^+ + \Sigma^-$

Step 2: Find I and I_3 for each particle

Kaon ($K^+ = u\bar{s}$):

- $I_3 = (+1/2) + 0 = +1/2$
- One light quark, so $I = 1/2$.

$$K^+ = \left|\frac{1}{2}, +\frac{1}{2}\right\rangle \tag{20}$$

Sigma ($\Sigma^- = dds$):

- $I_3 = (-1/2) + (-1/2) + 0 = -1$
- Part of the sigma triplet, so $I = 1$.

$$\Sigma^- = |1, -1\rangle \tag{21}$$

Direct-product ket: $|K^+\rangle|\Sigma^-\rangle = |1/2, +1/2\rangle|1, -1\rangle$.

Step 3: Build the total-isospin kets we need

Triangle rule: same as before, $1/2 \le I \le 3/2$, so $I = 1/2$ or $3/2$.
$I_3 = (+1/2) + (-1) = -1/2$. Same as the initial state. Good — conservation works!
We need the same kets: $|3/2, -1/2\rangle$ and $|1/2, -1/2\rangle$.

Step 4: We already decomposed these kets!

From the reaction (a) final state work, we already have Eq. 3 and Eq. 4:

$$\left|\frac{3}{2},-\frac{1}{2}\right\rangle = \sqrt{\frac{2}{3}}\,|K^0\Sigma^0\rangle + \sqrt{\frac{1}{3}}\,|K^+\Sigma^-\rangle \tag{Eq. 3}$$

$$\left|\frac{1}{2},-\frac{1}{2}\right\rangle = -\sqrt{\frac{1}{3}}\,|K^0\Sigma^0\rangle + \sqrt{\frac{2}{3}}\,|K^+\Sigma^-\rangle \tag{Eq. 4}$$

Now we need to isolate $|K^+\Sigma^-\rangle$ instead of $|K^0\Sigma^0\rangle$.

Step 5: Invert using systems of equations to get $|K^+\Sigma^-\rangle$

Goal: eliminate $|K^0\Sigma^0\rangle$ and solve for $|K^+\Sigma^-\rangle$.

Multiply Eq. 3 by $\sqrt{1/3}$:

$$\sqrt{\frac{1}{3}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle = \frac{\sqrt{2}}{3}\,|K^0\Sigma^0\rangle + \frac{1}{3}\,|K^+\Sigma^-\rangle \tag{3''}$$

Multiply Eq. 4 by $\sqrt{2/3}$:

$$\sqrt{\frac{2}{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle = -\frac{\sqrt{2}}{3}\,|K^0\Sigma^0\rangle + \frac{2}{3}\,|K^+\Sigma^-\rangle \tag{4''}$$

Add (3″) and (4″): The $|K^0\Sigma^0\rangle$ terms cancel!

$$\begin{aligned}\sqrt{\frac{1}{3}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle + \sqrt{\frac{2}{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle &= \left(\frac{1}{3}+\frac{2}{3}\right)|K^+\Sigma^-\rangle \\ &= 1\cdot|K^+\Sigma^-\rangle \end{aligned} \tag{22}$$

Therefore:

$$\boxed{|K^+\Sigma^-\rangle = \sqrt{\frac{1}{3}}\left|\frac{3}{2},-\frac{1}{2}\right\rangle + \sqrt{\frac{2}{3}}\left|\frac{1}{2},-\frac{1}{2}\right\rangle} \tag{Final-b}$$

Normalization check

$1/3 + 2/3 = 1$ ✓

Amplitude for Reaction (b)

Step 6: Compute the amplitude

Channel	Initial coeff.	Final coeff.	Product
$I = 3/2$	$\sqrt{1/3}$	$\sqrt{1/3}$	$\sqrt{1/3} \times \sqrt{1/3} = \frac{1}{3}$
$I = 1/2$	$-\sqrt{2/3}$	$\sqrt{2/3}$	$(-\sqrt{2/3}) \times (\sqrt{2/3}) = -\frac{2}{3}$

$$\mathcal{M}_b = \frac{1}{3}\,\mathcal{A}_{3/2} - \frac{2}{3}\,\mathcal{A}_{1/2} \tag{23}$$

$$\boxed{\mathcal{M}_b = \frac{1}{3}\left(\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2}\right)} \tag{24}$$

$$|\mathcal{M}_b|^2 = \frac{1}{9}\left|\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2}\right|^2 \tag{25}$$

Reaction (c): $\pi^+ + p \to K^+ + \Sigma^+$

Step 1: Write Out the Reaction

$$\pi^+ + p \longrightarrow K^+ + \Sigma^+ \tag{26}$$

This reaction has a different initial state from (a) and (b), so we start fresh.

Initial State: $\pi^+ + p$

Step 2: Find I and I_3 for each particle

Pion ($\pi^+ = u\bar{d}$):

- $I_3 = (+1/2) + (+1/2) = +1$
- Pion triplet, so $I = 1$. (Also: $I_3 = +1$ only fits in $I \geq 1$, and it's a stretched state.)

$$\pi^+ = |1, +1\rangle \tag{27}$$

Proton: same as before, $p = |1/2, +1/2\rangle$.
Direct-product ket: $|\pi^+\rangle|p\rangle = |1, +1\rangle|1/2, +1/2\rangle$.

Step 3: Build the total-isospin kets we need

Triangle rule: $1/2 \leq I \leq 3/2$, so $I = 1/2$ or $3/2$.
$I_3 = (+1) + (+1/2) = +3/2$.
Which total-isospin states have $I_3 = +3/2$?

- $I = 3/2$: allowed I_3 goes up to $+3/2$. **Yes.**
- $I = 1/2$: maximum I_3 is $+1/2$. **No!** $+3/2$ doesn't fit.

Only one ket survives: $|3/2, +3/2\rangle$.

Step 4: Decompose — but it's trivial!

The only product state with $m_1 + m_2 = +3/2$ is $m_1 = +1, m_2 = +1/2$. That's $|\pi^+\rangle|p\rangle$ itself. The CG table entry is just 1.

$$\left|\frac{3}{2}, +\frac{3}{2}\right\rangle = |\pi^+\rangle|p\rangle \tag{Eq. 5}$$

There's nothing to invert — it's already a single product state.

Step 5: Inversion is trivial

$$\boxed{|\pi^+ p\rangle = \left|\frac{3}{2}, +\frac{3}{2}\right\rangle} \qquad \text{(Initial-c)}$$

Pure $I = 3/2$. This is a stretched state.

Final State: $K^+ + \Sigma^+$

Step 2: Find I and I_3

Kaon: $K^+ = |1/2, +1/2\rangle$ (same as before).
Sigma ($\Sigma^+ = uus$):

- $I_3 = (+1/2) + (+1/2) + 0 = +1$
- Sigma triplet, $I = 1$. (Also: stretched state, $I_3 = +1$ forces $I \geq 1$.)

$$\Sigma^+ = |1, +1\rangle \qquad (28)$$

Direct-product ket: $|K^+\rangle|\Sigma^+\rangle = |1/2, +1/2\rangle|1, +1\rangle$. Total $I_3 = +1/2 + 1 = +3/2$.

Steps 3–5: Same stretched state argument

$I_3 = +3/2$ only fits in $I = 3/2$. Only one product state has $M = +3/2$. The CG coefficient is 1.

$$\boxed{|K^+\Sigma^+\rangle = \left|\frac{3}{2}, +\frac{3}{2}\right\rangle} \qquad \text{(Final-c)}$$

No systems of equations needed — stretched states are automatic.

Amplitude for Reaction (c)

Step 6: Compute the amplitude

Channel	Initial coeff.	Final coeff.	Product
$I = 3/2$	1	1	$1 \times 1 = 1$
$I = 1/2$	0	0	0

$$\boxed{\mathcal{M}_c = \mathcal{A}_{3/2}} \qquad (29)$$

$$|\mathcal{M}_c|^2 = |\mathcal{A}_{3/2}|^2 \tag{30}$$

Key observation about reaction (c)

Since $|\pi^+ p\rangle$ is pure $I = 3/2$, it has **zero overlap** with the $I = 1/2$ amplitude. This means:

- When $I = 3/2$ dominates: reaction (c) is at full strength.
- When $I = 1/2$ dominates: reaction (c) is **completely forbidden** ($\sigma_c = 0$).

Taking the Ratios

Collecting all amplitudes

$$\mathcal{M}_a = \frac{\sqrt{2}}{3}\left(\mathcal{A}_{3/2} + \mathcal{A}_{1/2}\right) \quad \Longrightarrow \quad |\mathcal{M}_a|^2 = \frac{2}{9}\left|\mathcal{A}_{3/2} + \mathcal{A}_{1/2}\right|^2 \tag{31}$$

$$\mathcal{M}_b = \frac{1}{3}\left(\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2}\right) \quad \Longrightarrow \quad |\mathcal{M}_b|^2 = \frac{1}{9}\left|\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2}\right|^2 \tag{32}$$

$$\mathcal{M}_c = \mathcal{A}_{3/2} \quad \Longrightarrow \quad |\mathcal{M}_c|^2 = \left|\mathcal{A}_{3/2}\right|^2 \tag{33}$$

Cross section is proportional to $|\mathcal{M}|^2$.

Case 1: $I = 3/2$ dominates (set $\mathcal{A}_{1/2} = 0$)

$$\sigma_a \propto \frac{2}{9}|\mathcal{A}_{3/2}|^2 \tag{34}$$

$$\sigma_b \propto \frac{1}{9}|\mathcal{A}_{3/2}|^2 \tag{35}$$

$$\sigma_c \propto |\mathcal{A}_{3/2}|^2 \tag{36}$$

$|\mathcal{A}_{3/2}|^2$ cancels in the ratio. Multiply through by 9:

$$\boxed{\sigma_a : \sigma_b : \sigma_c = 2 : 1 : 9} \tag{37}$$

Case 2: $I = 1/2$ dominates (set $\mathcal{A}_{3/2} = 0$)

$$\sigma_a \propto \frac{2}{9}|\mathcal{A}_{1/2}|^2 \tag{38}$$

$$\sigma_b \propto \frac{1}{9} \cdot 4\,|\mathcal{A}_{1/2}|^2 = \frac{4}{9}|\mathcal{A}_{1/2}|^2 \tag{39}$$

$$\sigma_c \propto 0 \tag{40}$$

$|\mathcal{A}_{1/2}|^2$ cancels. Multiply through by $9/2$:

$$\boxed{\sigma_a : \sigma_b : \sigma_c = 1 : 2 : 0} \tag{41}$$

Final Answer

Summary

	Amplitude $\mathcal{M}$	$I = 3/2$ dom.	$I = 1/2$ dom.
(a) $\pi^- p \to K^0\Sigma^0$	$\frac{\sqrt{2}}{3}(\mathcal{A}_{3/2} + \mathcal{A}_{1/2})$	2	1
(b) $\pi^- p \to K^+\Sigma^-$	$\frac{1}{3}(\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2})$	1	2
(c) $\pi^+ p \to K^+\Sigma^+$	$\mathcal{A}_{3/2}$	9	0

Consistency check

$$\mathcal{M}_c = \sqrt{2}\,\mathcal{M}_a + \mathcal{M}_b \tag{42}$$

$$\begin{aligned}\sqrt{2}\cdot\frac{\sqrt{2}}{3}(\mathcal{A}_{3/2} + \mathcal{A}_{1/2}) + \frac{1}{3}(\mathcal{A}_{3/2} - 2\mathcal{A}_{1/2}) &= \frac{2}{3}\mathcal{A}_{3/2} + \frac{2}{3}\mathcal{A}_{1/2} + \frac{1}{3}\mathcal{A}_{3/2} - \frac{2}{3}\mathcal{A}_{1/2} \\ &= \mathcal{A}_{3/2} = \mathcal{M}_c \quad \checkmark \end{aligned} \tag{43}$$

The recipe you followed for each reaction

1. **Write the reaction.**
2. **Find** I **and** I_3 for each particle (from PDG or stretched state argument). Write their kets $|I, I_3\rangle$.
3. **Triangle rule** on the state you're working with → find allowed total I values.
4. I_3 **conservation** → narrow down to which $|I, I_3\rangle$ kets you actually need.
5. **Decompose those kets** using the CG table: read the row for each $|I, I_3\rangle$ to write it in terms of direct-product kets. (Apply coupling-order phase if particles are in the "wrong" order for the table.)
6. **Invert with systems of equations**: you now have two equations (the decompositions) with two unknowns (the product kets). Multiply by the right factors and add to isolate the product ket you want. This gives you the reaction's product state in terms of total-isospin kets.
7. **Compute the amplitude**: multiply initial coefficient × final coefficient for each I channel, sum up.
8. **Square and take ratios**: $\sigma \propto |\mathcal{M}|^2$.